PRECLINICAL FOLLOW-UP PROGRAM

Care for today, prepare for tomorrow

Follow-up program for asymptomatic carriers of mutations in the prion protein gene

Having information about your genetics is an act of courage. This website is designed to accompany you: to explain what it means to be a carrier, why regular follow-up is a powerful tool, and what opportunities the near future holds

01

Understanding your starting point

What it means to be a carrier

You are a carrier of a genetic variant (also called a mutation, a change in the DNA compared with most of the population) that can be associated with the development of a hereditary prion disease. This means there is an increased risk, not a certainty. Many people with this genetic condition live decades without developing any symptoms, and science is advancing rapidly toward preventive treatments.

Being a carrier is not the same as having the disease. It is a condition that gives you valuable information — and that information, used well, can become a real advantage.

- **Not all carriers develop symptoms.** Penetrance —that is, the probability that the mutation will end up causing the disease— is high, although the age of onset varies considerably depending on the specific mutation and other individual factors that we still do not understand.

- **Knowledge is power.** Knowing your genetic status lets you make informed decisions, plan for the future, and access specialized follow-up programs.

- **Science isn't standing still.** There has never been more active research into prion diseases than there is right now. We are at a decisive moment for the development of preventive therapies.

02

Why follow-up matters

Four reasons to join a preclinical follow-up program

Follow-up programs for asymptomatic carriers are not just a research and medical monitoring tool. They are an active-preparation platform — for you and for all those who will come after – looking toward the preventive therapies being developed.

Early detection

Early detection: if subtle biological changes were to occur, they would be identified at the most opportune moment (before symptoms appear) so that available tools can be used.

Priority access

Priority access: participants in follow-up programs would be the best-positioned individuals to access new preventive therapies when clinical trials open their doors, since more information would be available about them than about other carriers.

Real information

Regular follow-up provides objective data about your brain health, replacing uncertainty with grounded knowledge. It is information you can access — or choose not to know yet.

Collective contribution

Your participation helps us better understand the disease and speeds up the development of treatments for you and for future generations.

03

Two realities within the same profile

Not all asymptomatic carriers are at the same moment

Within the group of asymptomatic carriers, current research makes it possible to identify two clearly distinct situations, even though in both the person remains without clinical symptoms:

Carrier in an inactive phase

The biological process associated with the mutation shows no signs of activity detectable with current techniques. This is linked to a longer time horizon before the first symptoms may appear.

Carrier in a silent active phase

There is incipient biological activity — without clinical symptoms— that specialized techniques can begin to detect. This profile is linked to a shorter time horizon, although we still do not know exactly how much shorter, or whether there is significant variability between carriers from the detection of biological activity to the appearance of the first signs. Even so, it is the group most likely to gain access to the first preventive clinical trials.

Finding out which situation you are in is voluntary. The follow-up program will offer you the possibility of knowing, always with support and without pressure.

04

What's on the horizon

Therapies in development and clinical trials

There have never been so many therapeutic strategies active at the same time in prion diseases. Several have already shown promising results in animal models, and some have begun to make the leap to human trials. This does not guarantee that all will work in humans, but it does mark a historic shift: for the first time, several real pathways are advancing in parallel toward future preventive and disease-modifying treatments.

WHAT EACH LABEL MEANS

Clinical trial underway

Already being tested in people.

Clinical trial registered, pending start

The study is already defined and authorized/registered, but has not yet started recruiting or administering the treatment.

Preclinical development

The therapy has shown activity in the laboratory and/or in animals, but has not yet been tested in people.

Clinical trial underway (phase 1/2a)

Antisense oligonucleotides (ASOs)

These are molecules designed to reduce the production of the prion protein. In animal models, lowering this protein has delayed disease onset and extended survival. This strategy is already being evaluated in people with symptomatic prion disease in an international clinical trial.

Clinical trial registered, pending start (phase 1)

RNA interference (siRNA)

These molecules “silence” the genetic message that gives rise to the prion protein. In animals they have achieved deep and lasting reductions of this protein, with beneficial effects on disease progression. In addition, a first phase 1 clinical trial has already been registered for this strategy, although recruitment has not yet begun.

Preclinical development

Small molecules for oral administration

Compounds are being developed that can reduce the amount of prion protein the cell manages to produce. Their great potential advantage is that they could be given as simple, long-term treatments. The public data reported so far show promising results in animals, but this approach has not yet moved on to clinical trials.

Preclinical development

Gene therapies

These include long-lasting silencing strategies and molecular-editing approaches aimed at stably switching off or reducing the prion protein gene. In animal models, they have already achieved a broad reduction of the protein and a significant extension of survival. Some programs are moving toward the clinic, but for now they remain in the preclinical phase.

Being enrolled in a follow-up program puts you on the front line to access these trials, should they be conducted in Spain. The window of opportunity is opening — and those already in follow-up will have the best chance of being among the first to cross it.

05

What the follow-up includes

The periodic visit protocol

The program consists of periodic visits — usually yearly or every two years — at a specialized center. Each visit is designed to be as comfortable and efficient as possible, always with the support of a dedicated team.

Neurological and neuropsychological evaluation

A detailed clinical examination and a battery of cognitive tests to establish and monitor your individual baseline.

Biomarkers in blood, cerebrospinal fluid, and skin

Blood tests and, when indicated, lumbar puncture and/or skin microbiopsy to measure specific proteins that reflect the neurobiological state.

Sleep study

Using polysomnography, we record what happens during sleep: movements, breathing, brain activity, and behavior. Certain sleep alterations – particularly those related to REM phase – can appear especially in carriers of Fatal Familial Insomnia, subtly before the first symptoms appear.

High-resolution imaging

Brain MRI that can detect structural or functional changes before they are clinically evident.

Psychological support

Emotional accompaniment and specialized guidance, available at every visit and accessible between visits when needed.

Updates on new therapies

Periodic updates on the clinical trials that are available or being prepared, for which you may be eligible, with personalized guidance for each opportunity.